The Foundation: $(\mathbb{S}, \tau, \omega)$ and Unsharp Unity

Opening statement for the next series. Bare minimum only.

Revision 10.

---

1. The triple

[DEF] $\mathbb{S} = \mathbb{R}_\varepsilon[(\mathbb{Z}_2)^4]$: the sixteen-dimensional real Cayley–Dickson algebra — the specific cocycle produced by iterated doubling from $\mathbb{R}$, not an arbitrary twisted group algebra on $(\mathbb{Z}_2)^4$ (Clifford algebras arise from other cocycles on the same group and are not meant here). Unital, flexible, quadratic, power-associative, central simple, positive-definite norm — and, stated with the same weight, **not** associative, **not** alternative, and possessed of zero divisors: these failures are starting data too, not omissions from a list of virtues. $\mathrm{Aut}(\mathbb{S}) = G_2 \times S_3$.

[DEF] τ: an involution in the S_3 factor. $\mathbb{O} = \mathrm{Fix}(\tau)$, $\mathbb{O}e_8 = \ker(\tau + 1)$. Three conjugate splits exist, related by the order-three μ; the choice among them is symmetric — an orbit coordinate, not a further commitment.

[FACT] The mediator direction used to build X, P, W below is, by the same logic, an orbit coordinate: G_2 acts transitively on the unit sphere in $\mathrm{Im}(\mathbb{O})$, so any choice of mediator is related to any other by an automorphism, and none is a further commitment.

[FACT, resting on a cited construction, not independently audited here] A dilation symmetry absent from the parent octonions is created by a specific contraction on $\mathrm{Im}(\mathbb{O}) = \mathrm{Im}(\mathrm{Fix}(\tau))$, survives as an automorphism through the tower that follows, and its cited unique $\hbar$-preserving component is the squeeze flow used throughout. The mediator's freedom is genuine and shown above; the contraction scheme itself — its grading and scaling regime — is asserted forced by the cited construction, not re-derived here. This is the single citation on which the inventory claim below, §2's family, and χ's entire domain most directly depend; it is the first item on the ledger.

[DEF] ω: a regular quasi-free state, normal on its own GNS type III_1 factor — a generic member of a sector family rather than one fixed point. Which member is physically realized is physical data, not part of the formal starting point. The construction of the operational algebra, and the results cited in §2, are established in a specific, named prior body of work: the second-contraction paper (the $\mathfrak{so}(3)$ reduction and the flow above), the squeeze-thermal-state paper (the type-I no-go, the scale-line construction, the type III_1 verdict, hyperfiniteness), and the centralizer paper (closing the last construction-specific link). None of it is re-derived here; all of it is cited, by title rather than by locator. The checking done on these results within this collaborative process is real but is not independent peer review or formal proof-checking; extracting the type-III_1 proof into a standalone, independently checkable artifact is the highest-priority companion document.

[SCOPE P-Em] The order of deduction is not the order of being. $\mathbb{S}$ is the persistence-form of a ground outside all algebraic and metric vocabulary; ω is that ground's sole imprint — one kind of object mediates the ground's presence, not one formally-singled-out instance of it; which member of the family is realized is empirical input, not formal data. Every claim below is scoped *among representable structures*. This does no deductive work — nothing is derived from it. Its function is to scope every other claim in this document.

$(\mathbb{S}, \tau, \omega)$ is the whole of the formal beginning. No further formal data is added; the realized sector is empirical input.

---

2. Unsharp unity

Two claims, of different logical kinds, both necessary, neither to be confused with the other.

[FACT] The $\mathfrak{so}(3)$ remnant arises in two steps on $\mathbb{O} = \mathrm{Fix}(\tau)$, the octonion algebra where compositionality genuinely holds, acting through its seven-dimensional imaginary part $\mathrm{Im}(\mathbb{O})$: fixing the mediator direction alone gives $\mathfrak{su}(3)$, the classical stabilizer of a point on S^6 ($G_2/SU(3) = S^6$); additionally fixing the P-triple as a real subspace compatible with the resulting complex structure reduces this to $\mathfrak{so}(3)$. The reduction and its dimensions rest on a symbolic argument — derivations of a composition algebra are skew with respect to the norm form — with machine-precision checks in the same paper confirming specific instances on top of it, not in place of it. This same $\mathfrak{so}(3)$ survives as an exact automorphism through the entire contraction tower. That it preserves ω itself, rather than merely mapping invariant states to invariant states, follows from ω's proven uniqueness among regular β-KMS states at its parameters, provided the relevant regularity class is stable under the $\mathfrak{so}(3)$ action — expected for Bogoliubov automorphisms, not separately verified here.

[PROP, proven within any established folium; open across the family] No admissible configuration is flowless: by the freeze dichotomy, flow trivial $\iff$ tracial, and no normal tracial state exists — this holds for every state normal to a folium of established type, in particular ω's own. No trace exists at all, not even a normal semifinite one, on the folium's von Neumann algebra; its modular spectrum locks the type specifically to III_1. (Non-faithful normal states are covered via their support corners, which remain type III.) **Open**: whether this type verdict holds uniformly across every member (r, β) of the family below, or has been established only at ω's own parameters — parameter-uniformity is its own ledger entry, not folded into this proposition.

[DEF] The symmetric reference is the pure vacuum of the unsqueezed, zero-temperature configuration — the corpus's own earlier language for the unity horizon, "the free scaffold vacuum." It is a factor state, as is any genuinely thermal member of the family; for factor states, quasi-equivalence and disjointness are the only two possibilities. Its disjointness from any thermal member needs no covariance estimate: quasi-equivalent representations generate isomorphic von Neumann algebras, and isomorphism preserves type; the reference, being pure, generates a type I folium ($B(H)$, acting irreducibly). Disjointness from ω's own folium is accordingly unconditional, since that folium's type is established directly. Disjointness from every other family member holds too, but currently inherits the same conditionality as the [PROP] above — pending only that the member's folium is *not type I*, which is strictly weaker than pinning III_1 specifically: any non-type-I folium already forbids isomorphism with $B(H)$, hence forbids quasi-equivalence, hence gives disjointness by the dichotomy, without needing the full Connes invariant. This decomposes the parameter-uniformity ledger item into a cheap half (not type I, sufficient here) and an expensive half (the full III_1 verdict, needed elsewhere for the modular facts) rather than charging this exclusion for the whole thing.

[PROP, proven, the endpoints distinguished] The operational algebra is CCR/Weyl, not CAR. Bosonic quasi-free covariance satisfies $\sigma \geq \frac{1}{2} \cdot \mathrm{Id}$, with equality exactly at the pure, unsqueezed vacuum — $\frac{1}{2}$ marks the *symmetric* end, not the settled one. The trace is not a quasi-free state at all: it is the non-regular counting trace on the Weyl algebra, with no covariance operator and no continuous field operators in its GNS representation. The two endpoints are distinct for a categorical reason, not a spectral one — one is a regular quasi-free state, the other is not in that class to begin with.

[FACT] States differing in global inverse temperature within a common squeeze frame are unconditionally disjoint: extremal KMS states of the same dynamics at distinct temperatures are mutually disjoint — a standard structural theorem (Bratteli–Robinson, *Operator Algebras and Quantum Statistical Mechanics II*, §5.3, "KMS-States"; the specific theorem number within that section is not pinned here), not a computation, and independent of any covariance estimate. The theorem's clean form presumes the states themselves are not tracial: for nontrivial dynamics, no state is KMS at two distinct temperatures, and tracial states are exactly the edge case that sits at the trivial-dynamics degeneracy, KMS at every β at once. That proviso is discharged automatically by the [PROP] above, which already excludes normal tracial states from these folia; the no-trace fact is what licenses the very disjointness theorem χ's well-definedness will lean on. Across squeeze frames, the states are KMS for conjugated rather than identical flows, and that theorem does not directly apply; there, disjointness rests on the covariance difference failing to be Hilbert–Schmidt, which the per-mode differences $\frac{1}{2}[\coth(\beta\omega_k/2) - \coth(\beta'\omega_k/2)]$ vanishing at high frequency make a question about mode density at bounded frequency — the same spectral input the [TARGET] below already owes, not a separate debt. Across the full (r, β) grid more generally, disjointness is only generic — a coincidental match between a squeeze-and-temperature pair and a different one is not ruled out here. The squeeze-thermal family is accordingly not a set of normal states of one von Neumann algebra; each member generates its own type III_1 folium.

[DEF] Admissible configuration means any state normal to *some* family member's folium — the union of folia, with the quasi-free family serving as a transversal that parametrizes them, not as the whole of what is admissible.

[DEF, χ as sector label] χ extends from the family to the full union by constancy on each folium: distinct values mark disjoint GNS representations, not distinct expectation values within one. This extension's well-definedness is currently conditional, not established outright: it is unconditional within a common squeeze frame, on the strength of the KMS-disjointness theorem above — distinct β at fixed squeeze guarantees distinct folia, which is what constancy-per-folium requires there. Across squeeze frames, distinct-β pairs are exactly the case the [TARGET] below has not yet settled; until the mode-density question is resolved, χ's well-definedness across frames is inherited from that open computation, not separately secured. The generic case (equal β, different squeeze) is harmless regardless, since χ agrees on those points whether or not the folia coincide. Connes–Størmer homogeneity holds within a folium; it is precisely why a genuine order parameter has to distinguish between them, not a threat to χ's existence.

[DEF, χ] χ is read from the *symplectic spectrum* of the covariance operator, not the covariance itself. Symplectic eigenvalues are invariant under every symplectic transformation, squeezing included: every point of the pure stratum — any squeeze, zero temperature — collapses to the single value $\nu = \frac{1}{2}$, and thermal covariance gives $\nu(\beta) = \frac{1}{2}\coth(\beta\omega/2)$ per mode, strictly between $\frac{1}{2}$ and ∞, independent of squeeze. The pure stratum is one boundary value; the non-regular tracial stratum is the $\beta \to 0$ limit; the family's closure adds both without ever containing either. **[TARGET]** The one piece not yet settled is uniformity in the thermodynamic limit: if the scale-line spectrum reaches $\omega_k \to 0$, the per-mode invariant diverges there at every temperature, and the correct global invariant is per-mode or spectral-density-weighted rather than a bare supremum. This same mode-density data settles the cross-squeeze-frame case of the disjointness fact above, and with it, cross-frame well-definedness of χ as a sector label — one target, carrying two stakes rather than one.

[FACT] Because χ is built from symplectic eigenvalues, it is exactly conserved by any symplectically implemented dynamics whatsoever — the derived squeeze flow included, for every state it transports, not only at equilibrium. This is the correct signature of a legitimate sector label, not a defect of the definition. It has a consequence worth stating outright: settling — any actual change in χ — is necessarily *non-Bogoliubov*, dissipative or open-system or restriction-based, and no such mechanism is currently in this document's inventory. The derived flow is, at most, the dynamics internal to a fixed sector; it is not, and structurally cannot be, the mechanism that moves between sectors.

[FACT] χ is coarse throughout the family, not only on the pure stratum: sectors at equal temperature but different squeeze share one χ value while remaining mutually disjoint. χ labels settling depth, not sector identity.

[OPEN] $\mathrm{Aut}(\mathbb{S}) = G_2 \times S_3$; the above concerns only the G_2-descended remnant. μ acts on the same Weyl C*-algebra. If μ is symplectically implemented — the same regularity-stability proviso as for $\mathfrak{so}(3)$ — then $\chi \circ \mu = \chi$ follows automatically from the conservation fact above, and the live content of this question narrows to whether μ permutes sectors* within* a single χ-fiber or fixes each one. More generally, this is one instance of a question covering every orbit coordinate §1 declares free — the split τ and the mediator direction alike: does the settled configuration select a specific representative, physically, of a symmetry the formalism treats as free? If μ permutes χ-equal sectors, the settled configuration selects a specific split, and §1's "orbit coordinate, not a further commitment" holds only formally, not physically — and the same question, unresolved here, applies identically to the mediator's own orbit-coordinate status.

[POST, motivated, not proven] The near-symmetric end is placed, specifically, toward the direction later identified as the past — not the future, not both alike; a zero-parameter boundary condition, with precedent in the Weyl-curvature-hypothesis tradition, naming a choice between two provably distinct ends. Global sectors are disjoint, but disjoint sectors are typically *locally* quasi-equivalent — local normality is the standard situation — so restriction to local subalgebras of a net is not merely a resource beyond the bare triple; it is the canonical place cross-sector comparison, and with it any monotonicity of χ, becomes possible at all. Given the conservation fact above, it is also the canonical place *any* mechanism of settling would have to live, since the derived flow structurally cannot supply one.

[SCOPE] This section is bosonic throughout. A successful fermionic construction (§3) adds a CAR sector, and fermionic endpoint geometry runs opposite to bosonic — the CAR trace is a regular quasi-free state interior to the covariance range, with pure states at the extremes. On a combined algebra, "the settled end" is not the single thing described here; that reopening is a declared dependency of §3 succeeding, not a present defect.

---

3. The seed, stated as a commitment

[FACT] No nonzero real element of $\mathbb{S}$ can supply a nilpotent structure by serving *directly* as the Clifford/CAR product — shown independently by the algebra's positive-definite norm and by a general trace identity on its left-multiplication operators. This is not the obstacle a fermionic construction actually faces: the standard route (self-dual CAR, the same family Majorana fermions are built from) needs only Clifford relations over a real quadratic space, which $\mathbb{S}$'s positive-definite norm supplies unconditionally, once the Clifford product is built abstractly from the quadratic form rather than borrowed from $\mathbb{S}$'s own multiplication.

[REFRAMED] The open question is where the *polarization* comes from — the choice of complex structure turning the abstract, unconditionally-existing real Clifford algebra into a physical Fock representation with genuine, nilpotent, particle-like creation operators. Nilpotency is a feature of the chosen representation, not of the underlying algebra. Live candidates: a state-selected dynamics (this document's working choice); a seed complexified from the outset; the zero-divisor structure of $\mathbb{S}$ itself; or declining a particle interpretation and working with the self-dual algebra directly.

[COMMITMENT] $\mathbb{S}$ is kept real; the polarization is expected to come from ω rather than be adjoined from outside. This is motivated by a mechanism already working for the bosonic sector: a real classical system, a derived flow, a Kay–Wald one-particle structure yielding a genuine complex Hilbert space, no adjunction. Once a state exists, Tomita–Takesaki theory supplies its own modular flow, guaranteed for any faithful normal state and certifying nothing further by itself; whether that flow agrees with the dynamics used to build the one-particle structure is the KMS property, verified per construction — what the bosonic sector's own proof supplies, not a general consequence of Tomita–Takesaki. The state-selected route is the working choice on precedent, not on necessity.

[FACT] If the operational algebra lands, as intended, on the hyperfinite type III_1 factor, Haagerup's uniqueness theorem means its bare isomorphism class carries no memory of $\mathbb{S}$, τ, or the specific cocycle — among *hyperfinite* such factors, every one is the same algebra up to isomorphism (non-injective type III_1 factors, such as the free Araki–Woods factors, are different objects, not covered here). Everything distinctive lives in the decorations: the state ω and the sector structure it belongs to, the $\mathfrak{so}(3)$ remnant's action, and the subfactor generated by $\mathrm{Fix}(\tau)$.

[FACT] Where $\mathbb{S}$'s actual multiplication does irreplaceable work — not its automorphism group, not its norm alone, since every 16-dimensional positive-definite form is isomorphic and carries no sedenion-specific content by itself: $J = L_{e_4}$, left multiplication by the mediator direction, serves directly as the complex structure in the $\mathfrak{so}(3)$-remnant derivation. This is one confirmed instance, not a complete inventory.

[CONDITION OF REVISION] This commitment holds provided the polarization can be supplied by a state-selected dynamics — a real Clifford structure over $\mathbb{S}$'s own quadratic form, a state-selected one-particle structure, matched against the published algebraic construction of fermion content. That construction is not complete. If it is shown that no such polarization reproduces the required content, the live alternatives above — a complexified seed foremost — become the honest fallback.

---

4. What this document is

A statement of starting data and its status, nothing more. It proves nothing about gauge structure, mass, or cosmology, and claims nothing beyond what is marked proven. Its only obligation is that every sentence carry its correct modal weight — proposition, fact, postulate, or commitment — so that what follows can be built without inheriting a hidden assumption it was never told it was making.

[LEDGER] Currently open, roughly in order of priority: exact section and theorem locators for the contraction-scheme paper specifically, now the keystone citation for §1's inventory claim, §2's family, and χ's domain jointly; the mechanism of settling itself — necessarily non-Bogoliubov, not currently supplied anywhere in this inventory; the thermodynamic-limit uniformity of χ's symplectic invariant, now doing double duty as the cross-frame well-definedness of χ itself (§2, [TARGET]); parameter-uniformity of the type III_1 verdict across the family (§2) — decomposed into a cheap half (not type I, sufficient for the reference's exclusion above) and an expensive half (the full III_1 verdict, needed for the modular facts elsewhere), rather than one undifferentiated item; whether μ permutes sectors within a χ-fiber (§2, [OPEN]); the direction postulate (§2, [POST]); the fermionic condition of revision (§3); the reopening of §2's endpoint geometry if §3 succeeds (§2, [SCOPE]); locators for the remaining three cited companion papers; the exact theorem number within Bratteli–Robinson §5.3 for the KMS-disjointness fact (§2); a complete accounting of where $\mathbb{S}$'s multiplication does irreplaceable work beyond $J = L_{e_4}$; and confirmation that the KMS-uniqueness regularity class is stable under the $\mathfrak{so}(3)$ action. This count changes as items resolve or new ones surface; it is carried here so the document is its own record.